

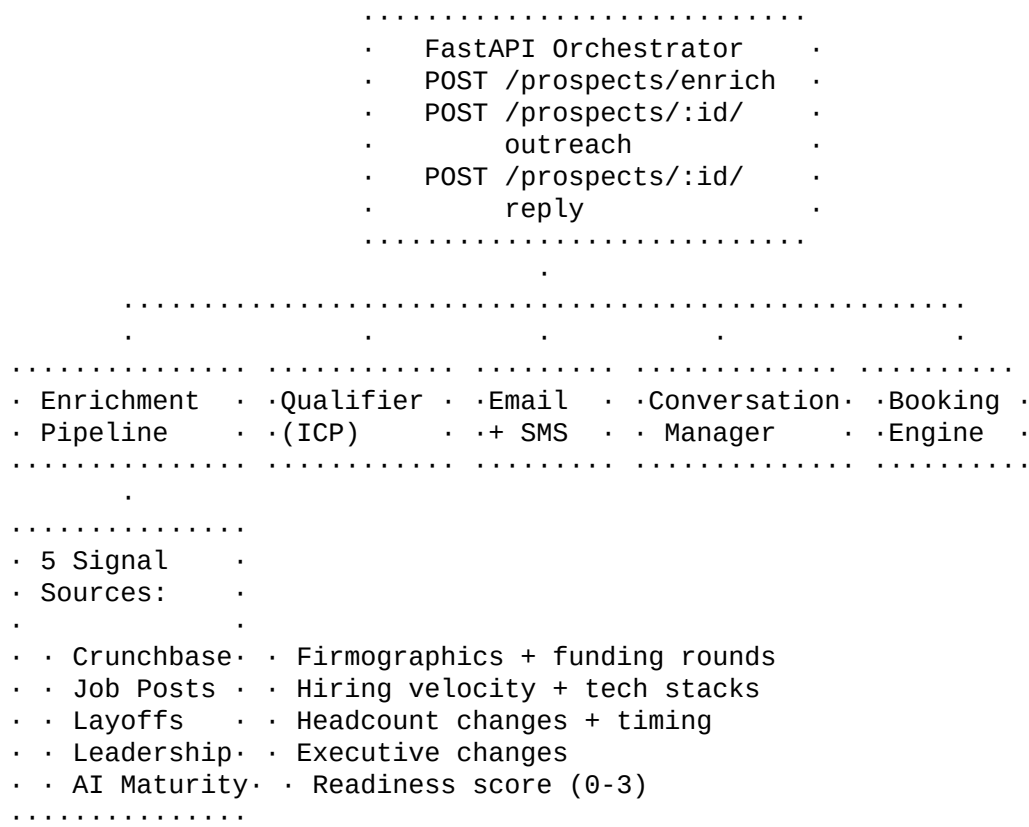
The Conversion Engine · Automated Lead Generation & Conversion System

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What It Does

End-to-end AI-powered system that enriches leads from raw company names into qualified prospects, composes personalized outreach, manages multi-turn conversations, and books discovery calls · all autonomously.

Architecture



Data Flow

- 1 **Enrich** · Runs 5 signal sources + competitor gap analysis · Classifies into ICP segment · Syncs to HubSpot CRM
 - 2 **Outreach** · Composes signal-grounded email using enrichment data + style guide · Sends via Resend
 - 3 **Reply** · Classifies reply intent (engaged/objection/hard/etc.) · Generates context-aware response · Updates CRM
 - 4 **Booking** · Cal.com integration for discovery call scheduling · Syncs to CRM
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Key Design Decisions

- **Centralized orchestrator** · state machine controls which actions fire and when
 - **Monotonic state transitions** · leads only move forward (NEW · EMAILED · REPLIED · QUALIFIED · BOOKED)
 - **Idempotent processing** · duplicate webhooks are harmless
 - **Kill switch** · all outbound routes to local sink until explicitly enabled
 - **Observability** · full Langfuse tracing on every LLM call
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Tech Stack

Python | FastAPI | OpenRouter (LLM) | Resend (email) | Africa's Talking (SMS) | HubSpot (CRM) | Cal.com (booking) | Langfuse (observability)

Results

- 5-signal enrichment pipeline producing qualified leads from raw company names
- Signal-grounded outreach (no hallucinated claims)
- Multi-turn conversation management with reply classification
- Full CRM sync with contact, company, deal, and note creation